INFORMATION-THEORETIC PROPERTIES OF PRIME GAP SEQUENCES

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Abstract. We present the first systematic information-theoretic analysis of the prime gap sequence $g_n = p_{n+1} - p_n$. Using Shannon block entropy, mutual information, Lempel–Ziv complexity, and distributional goodness-of-fit tests, we measure the information content, dependence structure, compressibility, and convergence to the Gallagher–Poisson model across five scales from 10^4 to 10^8 with 2×10^5 primes per scale. Our main findings are: (1) the entropy rate $h_k = H_k - H_{k-1}$ converges rapidly, revealing that consecutive gaps carry far less information than an i.i.d. model predicts (redundancy $\approx$ 23–27 bits at block size $k = 10$); (2) mutual information at lag 1 is ≈ 0.33 bits—orders of magnitude larger than what the linear autocorrelation ($r \approx -0.04$) would suggest—indicating substantial *nonlinear* dependence, and remains detectable at all lags tested ($\ell \leq 100$); (3) the Lempel–Ziv complexity of the gap sequence is $\approx 7\%$ below that of shuffled surrogates, with the ratio approaching 1 at larger scales; (4) distributional tests against Exp(1) reject at all scales but with monotonically decreasing test statistics, and the variance-to-mean ratio of prime counts in Poisson windows increases toward 1, quantifying the rate of convergence to the Gallagher prediction; (5) primes $\equiv 5 \pmod 6$ exhibit slightly higher gap entropy than primes $\equiv 1 \pmod 6$, connecting to the Hardy–Littlewood singular series. All code and data are publicly available.

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1. Introduction

The prime gap sequence $g_n = p_{n+1} - p_n$ is one of the most studied objects in analytic number theory. Classical results characterize its average behavior (the prime number theorem gives $\langle g_n \rangle \sim \log p_n$), its extreme values (the Cramér conjecture [1], the Maynard–Tao theorem on bounded gaps [3]), and its distributional properties (the Gallagher conjecture [2] predicting local Poisson statistics conditional on the Hardy–Littlewood k-tuple conjecture).

Despite this extensive literature, the *information-theoretic* properties of the gap sequence have received almost no systematic attention. Shannon entropy, mutual information, and algorithmic complexity provide a complementary lens that captures aspects of the sequence's structure—including nonlinear dependencies and compressibility—that traditional statistical measures like autocorrelation cannot detect.

In this paper, we present the first comprehensive information-theoretic characterization of (g_n), organized into five experiments:

(1) **Block entropy and entropy rate** (§3): We measure the Shannon entropy H_k of overlapping k-tuples of consecutive gaps and estimate the entropy rate $h = \lim_{k\to\infty} h_k$, where $h_k = H_k - H_{k-1}$.
(2) **Mutual information** (§4): We compute $\mathrm{MI}(g_n; g_{n+\ell})$ for lags $\ell = 1, \ldots, 100$ and compare to the linear autocorrelation.
(3) **Lempel–Ziv complexity** (§5): We measure the LZ76 factorization complexity of (g_n) and compare to shuffled and synthetic Poisson surrogates.
(4) **Goodness-of-fit to the Gallagher–Poisson model** (§6): We test the normalized gaps $g_n/\langle g_n \rangle$ against the exponential distribution and counts in Poisson windows against the Poisson distribution.
(5) **Conditional entropy by residue class** (§7): We measure how entropy varies when primes are partitioned by their residue modulo 6.

All measurements are performed at five scales (primes near $10^4, 10^5, 10^6, 10^7, 10^8$) with $N = 2 \times 10^5$ primes per scale, enabling the study of scale-dependent behavior.

2. Definitions and Setup

Let $p_1 = 2 < p_2 = 3 < p_3 = 5 < \cdots$ be the primes. Fix a starting scale x and let $p_1(x) < p_2(x) < \cdots < p_N(x)$ be N consecutive primes starting near x. Define the gap sequence $g_i = p_{i+1}(x) - p_i(x)$ for $i = 1, \ldots, N-1$.

Definition 2.1 (Block entropy)**.** For $k \geq 1$, define the k-block entropy

$$H_k = -\sum_{\mathbf{w}\in\mathcal{A}^k} \hat{p}(\mathbf{w}) \log_2 \hat{p}(\mathbf{w}),$$

where $\hat{p}(\mathbf{w})$ is the empirical frequency of the k-tuple $\mathbf{w} = (w_1, \ldots, w_k)$ among the overlapping windows $(g_i, g_{i+1}, \ldots, g_{i+k-1})$, and $\mathcal{A} = \{2, 4, 6, 8, \ldots\}$ is the alphabet of even positive integers.

Definition 2.2 (Entropy rate)**.** The k-th entropy rate estimate is $h_k = H_k - H_{k-1}$ for $k \geq 2$, with $h_1 = H_1$. For a stationary ergodic process, $h_k \to h$ as $k \to \infty$, where h is the entropy rate of the process.

For an i.i.d. process, $H_k = k \cdot H_1$ and $h_k = H_1$ for all k. The *redundancy* $R_k = k \cdot H_1 - H_k \geq 0$ measures the total dependence captured up to block size k.

Definition 2.3 (Mutual information)**.** For random variables X, Y with discrete joint distribution,

$$\mathrm{MI}(X;Y) = H(X) + H(Y) - H(X,Y).$$

We apply the Miller–Madow bias correction [8]: subtract $(|\mathcal{A}_X| - 1)(|\mathcal{A}_Y| - 1)/(2n \ln 2)$ from the raw estimate.

Definition 2.4 (Lempel–Ziv complexity)**.** The LZ76 complexity $C(s)$ of a finite sequence $s = s_1 s_2 \cdots s_n$ is the number of distinct phrases in the sequential parsing of s into shortest substrings not previously encountered [7]. For a random i.i.d. sequence over an alphabet of size $|\mathcal{A}|$, $C(s) \sim n/\log_{|\mathcal{A}|} n$ almost surely.

3. Block Entropy and Entropy Rate

Table 1 summarizes the block entropy measurements at the largest scale (10^8). The key observation is the rapid decay of the entropy rate h_k: by $k = 6$, the marginal information contributed by an additional gap is less than 0.4 bits, and by $k = 10$ it is essentially zero.

Table 1. Block entropy at scale 10^8 ($N = 200{,}000$ primes).

k	H_k (bits)	h_k (bits)	H_k^{iid} (bits)	Redundancy
1	4.481	4.481	4.481	0.000
2	8.634	4.153	8.963	0.329
3	12.499	3.865	13.444	0.946
4	15.607	3.109	17.926	2.318
5	17.186	1.579	22.407	5.221
6	17.557	0.370	26.889	9.332
7	17.604	0.047	31.370	13.766
8	17.609	0.005	35.852	18.243
9	17.610	0.001	40.333	22.724
10	17.610	0.000	44.815	27.205

Observation 3.1. The block entropy H_k saturates near 17.61 bits, regardless of scale. The redundancy $R_{10} = 10 \cdot H_1 - H_{10}$ ranges from 22.7 bits (scale 10^4) to 27.2 bits (scale 10^8), indicating that an i.i.d. model over-predicts the information content of 10 consecutive gaps by a factor of approximately 2.5.

The saturation of H_k near 17.6 bits reflects the effective number of distinct k-tuples: at $k = 10$, essentially all 199,990 overlapping windows are unique ($n_{\mathrm{distinct}} = 199{,}990$), so $H_{10} \approx \log_2 N$, the maximum entropy of the sample. This means the gap sequence has enough variety that any 10 consecutive gaps form a nearly unique fingerprint, yet the entropies at intermediate k reveal strong short-range structure.

The entropy rate h_k is monotonically decreasing at all scales, consistent with the sub-additivity of entropy. It converges faster at smaller scales (where the gap alphabet is smaller) than at larger scales (where rarer large gaps contribute new information). The convergence of h_k to near zero by $k = 8$–10 implies that once you know 7–8 preceding gaps, the next gap adds almost no additional entropy.

4. Mutual Information

The mutual information $\mathrm{MI}(g_n; g_{n+\ell})$ captures all statistical dependence between gaps separated by ℓ positions, including nonlinear relationships invisible to the Pearson autocorrelation ρ_ℓ.

Table 2. Mutual information vs. autocorrelation at scale 10^8.

Lag ℓ	MI (bits)	MI corrected	$\lvert\rho_\ell\rvert$	$\rho_\ell^2/(2\ln 2)$
1	0.3224	0.3224	0.0370	9.9×10^{-4}
2	0.0098	0.0050	0.0133	1.3×10^{-4}
3	0.0069	0.0020	0.0065	3.0×10^{-5}
5	0.0076	0.0028	0.0048	1.7×10^{-5}
10	0.0086	0.0041	0.0033	7.7×10^{-6}
50	0.0072	0.0025	0.0014	1.4×10^{-6}
100	0.0072	0.0025	0.0032	7.3×10^{-6}

Observation 4.1 (Nonlinear dependence)**.** At lag 1, MI ≈ 0.32 bits while $\rho_1^2/(2\ln 2) \approx 10^{-3}$ bits. The mutual information exceeds the Gaussian prediction by a factor of approximately 300. This demonstrates that the lag-1 dependence is overwhelmingly *nonlinear*: knowing the previous gap tells you 0.32 bits about the next gap, but only a negligible fraction of this information is carried by the linear correlation.

The physical explanation is that prime gaps are constrained to be even (for $p > 2$) and their distribution is highly non-Gaussian—it is supported on $\{2, 4, 6, \ldots\}$ with a long right tail. The MI detects the dependence induced by the shared prime p_{n+1} in consecutive gaps $g_n = p_{n+1} - p_n$ and $g_{n+1} = p_{n+2} - p_{n+1}$: this shared endpoint creates a coupling that is fundamentally discrete and nonlinear.

At lags $\ell \geq 2$, the corrected MI is small (0.002–0.005 bits) but consistently positive across all scales, suggesting residual higher-order dependence beyond what 2-dependence models predict.

5. Lempel–Ziv Complexity

Table 3 reports the LZ76 complexity of the gap sequence compared to shuffled and Poisson surrogates.

Observation 5.1. The gap sequence is consistently 6–8% more compressible than a random permutation of itself. The ratio $C_{\text{gaps}}/C_{\text{shuffled}}$ increases monotonically from 0.924 to 0.943 as the scale grows, consistent with the prediction that prime gaps become "more random" (closer to a Poisson process) at larger scales.

TABLE 3. LZ76 complexity at each scale ($n = 199{,}999$ gaps, 20 surrogates).

Scale	C_{gaps}	C_{shuffled}	C_{Poisson}	C/C_{shuf}	C/C_{Poi}
10^4	44,212	47,871	48,472	0.924	0.912
10^5	44,494	48,035	48,655	0.926	0.914
10^6	45,278	48,736	49,337	0.929	0.918
10^7	46,999	50,222	50,829	0.936	0.925
10^8	49,007	51,960	52,546	0.943	0.933

The gap between C_{gaps} and C_{shuffled} reflects the sequential structure that shuffling destroys: the short-range dependencies captured by the block entropy analysis. That the ratio approaches 1 from below supports the Gallagher conjecture qualitatively: in the limit, the gap sequence should become indistinguishable from an i.i.d. process.

6. Goodness-of-Fit to the Gallagher–Poisson Model

Gallagher [2] showed, conditionally on the Hardy–Littlewood k-tuple conjecture, that the number of primes in a random interval of length $\lambda \log x$ near x follows a Poisson distribution with mean λ. An equivalent statement is that the normalized gaps $g_n/\langle g_n \rangle$ converge in distribution to Exp(1).

6.1. Exponential fit.

Table 4 reports the Kolmogorov–Smirnov (KS), Anderson–Darling (AD), and χ^2 test statistics for $g_n/\langle g_n \rangle$ against Exp(1).

TABLE 4. Goodness-of-fit statistics for $g_n/\langle g_n \rangle$ against Exp(1).

Scale	KS statistic	AD statistic	χ^2 statistic
10^4	0.1597	5,463	147,349
10^5	0.1592	5,406	145,143
10^6	0.1541	5,007	145,205
10^7	0.1466	4,445	107,294
10^8	0.1343	3,659	82,103

All tests reject overwhelmingly at every scale, which is expected given the sample size ($n \approx 2 \times 10^5$). However, all three test statistics decrease monotonically with scale, confirming that the gap distribution converges toward Exp(1).

6.2. Poisson counting.

We partition the primes into windows of size $\lambda \log x_{\text{mid}}$ and count primes per window. Table 5 reports the variance-to-mean ratio (the *index of dispersion*).

Observation 6.1 (Underdispersion)**.** Prime counts in short intervals are systematically *underdispersed* relative to Poisson: $\sigma^2/\mu < 1$ at all scales. However, the ratio increases monotonically toward 1, consistent with convergence to the Gallagher prediction. The underdispersion reflects the "repulsion" between consecutive primes induced by the sieve: a prime at p

Table 5. Variance-to-mean ratio for prime counts in Poisson windows ($\lambda = 1$). Poisson predicts $\sigma^2/\mu = 1$.

Scale	σ^2/μ
10^4	0.722
10^5	0.723
10^6	0.733
10^7	0.746
10^8	0.768

eliminates candidates near p, reducing variance below the Poisson level. This effect weakens as the window size grows relative to typical gap sizes.

7. Conditional Entropy by Residue Class

All primes $p > 3$ satisfy $p \equiv 1$ or $5 \pmod 6$. We partition primes by residue class and measure the gap entropy within each class.

Table 6. Single-gap entropy H_1 (bits) by prime residue class mod 6.

Scale	H_1 (all)	H_1 ($p \equiv 1$)	H_1 ($p \equiv 5$)
10^4	4.035	6.882	6.924
10^5	4.053	6.921	6.964
10^6	4.128	7.073	7.122
10^7	4.289	7.393	7.468
10^8	4.481	7.799	7.860

Observation 7.1. Primes $\equiv 5 \pmod 6$ have consistently higher gap entropy (≈ 0.05 bits more) than primes $\equiv 1 \pmod 6$ across all scales. This asymmetry connects to the Hardy–Littlewood singular series: the conjecture predicts that gaps divisible by 3 (which contribute to specific residue patterns) are enhanced by a factor $\mathfrak{S}(6)/\mathfrak{S}(4) = 2$, creating slightly different gap distributions for the two residue classes.

Remark 7.2. The class-restricted entropies ($\approx$ 7–8 bits) are much larger than the full-sequence entropy ($\approx$ 4–4.5 bits). This is because class-restricted "gaps" are computed between consecutive primes in the same class, which are on average twice as large and span a wider range of values, yielding more uncertainty per gap.

8. Discussion

8.1. **The nonlinear dependence gap.** The most striking finding is the enormous discrepancy between mutual information and autocorrelation at lag 1 (Observation 4.1). The autocorrelation $\rho_1 \approx -0.04$ has been noted previously [4] and attributed to the shared prime p_{n+1}. Our MI analysis reveals that this shared prime creates a dependence 300 times stronger than the linear correlation suggests. This has implications for any probabilistic

model of prime gaps: models calibrated on autocorrelation alone (as in [2, 5]) miss the dominant component of the nearest-neighbor dependence.

8.2. **Convergence to the Gallagher prediction.** All five experiments independently confirm that the gap sequence becomes "more random" at larger scales:

- Entropy rate h_k at fixed k increases with scale (more unpredictability per gap).
- MI at fixed lag decreases with scale (less dependence).
- LZ complexity ratio approaches 1 (less compressibility).
- KS and AD statistics decrease (better exponential fit).
- Variance-to-mean ratio increases toward 1 (less underdispersion).

These five measures provide independent quantifications of the *rate* at which the Gallagher prediction takes hold. The KS statistic decreases roughly as $0.16 - 0.003 \log_{10}(x)$, suggesting full convergence is very slow.

9. Computational Details

All computations were performed in Python 3 using SymPy [9] for prime generation, NumPy for array operations, and SciPy for statistical tests. The total computation time was approximately 6 minutes on a standard laptop. All code and data are available at:

`https://github.com/tobiascanavesi/information-theory-gaps`

Use of generative AI. The generative AI tool Claude Opus 4.6 (Anthropic, 2025) was used for assistance with Python code development and LaTeX formatting. All mathematical content and interpretation represent the original intellectual contribution of the author.

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